

INC-2026-0716 - Order API Connection Pool Exhaustion During Flash Sale

Incident Report | Engineering

Document ID	INC-2026-0716	Seller ID	S002
Department	Engineering	Classification	CONFIDENTIAL
Owner	Vikram Singh	Created Date	2026-07-18
Related Scenario ID	SC-003	Document Type	Incident Report

1. Incident Summary

Field	Value
Incident ID	INC-2026-0716
Severity	SEV-1
Service	Order Service and API gateway
Component	Order placement path, inventory lookup dependency
Detected	2026-07-16 20:12 IST
Mitigated	2026-07-16 20:44 IST
Resolved	2026-07-16 21:05 IST
Incident owner	Vikram Singh, Engineering
Seller affected	S002 - BluePeak Electronics
Related ticket	TKT-2026-0716-268

During a BluePeak flash sale, order placement latency rose to a p99 of 12.4 seconds and 3,900 order placements failed with HTTP 503 over approximately 40 minutes. The order service connection pool was exhausted by an amplified retry load, worsened by the absence of a circuit breaker on a slow downstream inventory lookup.

2. Impact

Dimension	Detail
Failed order placements	3,900
Peak error rate	31 percent
Peak p99 latency	12.4 seconds, against a 900 ms target
Duration of significant degradation	approximately 40 minutes
Sellers affected	All sellers on the shared order path; S002 predominantly
Orders recovered by replay	1,120
Net lost orders	approximately 2,780
Commercial context	Flash sale with committed external media spend

This was classified SEV-1 rather than SEV-2 because the shared order placement path was degraded for all sellers, not only for the seller whose traffic triggered it.

3. Detection

Detected by the seller during the live event.

BluePeak's integration lead called Support at 20:07 IST, seven minutes after the sale opened, reporting that roughly one in three order placements was failing. Support confirmed from platform metrics and escalated at 20:12.

Platform latency alerts did fire, at 20:09, two minutes before escalation. The alert reached the on-call channel but the seller call had already started by then. In practice the seller and the alert detected the problem at effectively the same time, which is a poor outcome for a SEV-1 on a pre-announced event.

4. Timeline

Time (IST)	Event
16 Jul 20:00	Flash sale opens; traffic rises to approximately 9x baseline
16 Jul 20:02	Inventory lookup p99 begins rising
16 Jul 20:04	Order placement p99 exceeds 2 seconds
16 Jul 20:06	Client retries begin amplifying request volume
16 Jul 20:07	Seller calls Support
16 Jul 20:09	Latency alert fires to on-call
16 Jul 20:12	Escalated to Engineering; incident opened
16 Jul 20:16	Connection pool saturation observed at 120 of 120 in use
16 Jul 20:19	Pool exhaustion confirmed as the proximate cause
16 Jul 20:26	Emergency pool increase from 120 to 240 applied
16 Jul 20:31	Rate limiting applied to the highest-volume client
16 Jul 20:38	Error rate falls below 10 percent
16 Jul 20:44	Error rate below 2 percent; incident mitigated
16 Jul 21:05	Latency at baseline; sale continues without further failures
16 Jul 22:00	Sale ends
18 Jul 09:40	1,120 failed orders replayed where carts remained valid

5. Symptoms

- HTTP 503 responses from the order placement endpoint.
- p99 latency rising steadily rather than spiking, over roughly four minutes.
- Connection pool utilisation pinned at 100 percent.
- Inventory lookup dependency responding slowly but not failing.
- Total inbound request volume far exceeding the expected traffic profile for the announced sale size.

- CPU and memory on order service instances well within normal range.

The last symptom was important. The service was not resource-constrained. It was blocked, waiting on connections held by requests waiting on a slow dependency.

6. Affected Services

Service	Role	State
Order Service	Order placement	Degraded: pool exhausted
API gateway	Request ingress	Functioning; no load shedding configured
Inventory lookup service	Availability check during placement	Slow, not failed
Cart service	Cart retrieval	Unaffected
BluePeak client	Order placement caller	Amplifying load through unbounded retries

7. Investigation

7.1 The Proximate Cause

Order placement calls the inventory lookup service synchronously. Under normal load that call returns in about 40 ms. Under flash-sale load it slowed to 600-900 ms, because the lookup fans out across a much larger set of SKUs than a typical order.

Each in-flight placement holds a connection from a fixed pool of 120 for the duration of that call. At 9x baseline traffic with a 15x slower dependency, demand for connections exceeded 120 within about four minutes. Once the pool was full, new requests queued with no acquisition timeout and eventually returned 503.

7.2 The Amplification

The failure would have been self-limiting if clients had backed off. They did not.

BluePeak's client retried timed-out placements without limit and without backoff. Every 503 produced another request. Measured inbound volume at 20:18 was approximately 3.4 times the volume implied by actual buyer activity, so roughly 70 percent of the traffic at peak was retries of already-failed requests.

This is why no static capacity figure would have been sufficient. An unbounded retry loop expands to consume whatever capacity exists.

7.3 The Missing Control

The inventory lookup had no circuit breaker. A slow dependency should eventually be bypassed or failed fast, so that callers release their connections. Instead, every caller waited the full timeout, and the timeout on that path was 10 seconds.

A 10-second timeout with a 120-connection pool caps throughput at 12 requests per second once the dependency is slow, regardless of how much capacity the service otherwise has.

8. Root Cause

The order service connection pool was exhausted because synchronous inventory lookups slowed under flash-sale load and were not protected by a circuit breaker or an acquisition timeout, while an unbounded client retry policy amplified inbound traffic to roughly 3.4 times genuine demand. The combination converted a capacity pressure into a 40-minute cascading failure.

9. Contributing Factors

1. **No circuit breaker on a synchronous dependency** in the critical path.
2. **No connection acquisition timeout**, so requests queued instead of failing fast.
3. **Unbounded client retries** at the seller, with no backoff or jitter.
4. **HTTP 503 as the overload response**, which does not communicate a retry strategy. A 429 with Retry-After states the correct behaviour explicitly.
5. **No advance notice of the sale**. The platform learned of the event when the traffic arrived. No capacity assessment was possible.
6. **Pool sized for steady state**, never load-tested against a dependency degradation scenario.

10. Mitigation

Two actions restored service.

The emergency pool increase from 120 to 240 at 20:26 bought immediate headroom and reduced the error rate from 31 percent to approximately 10 percent within 12 minutes. This treated the symptom.

Rate limiting the highest-volume client at 20:31 addressed the amplification and took the error rate below 2 percent by 20:44. This was the action that actually ended the incident, and it is uncomfortable: the platform throttled a seller's traffic during that seller's own revenue event. It was the correct call, because the throttled traffic was overwhelmingly retries rather than genuine buyer demand.

Time from escalation to mitigation: 32 minutes.

11. Resolution

Permanent remediation was delivered under the S002 throughput and resilience programme, workstream WS-A, on 26 August 2026:

- Circuit breaker on the inventory lookup, opening at a 25 percent error rate over a 10-second window with a 30-second half-open probe.
- Connection pool resized to 300 with an explicit acquisition timeout of 750 ms, so a saturated pool fails fast.
- Request-level timeout of 2.5 seconds on the order placement path, replacing 10 seconds.
- Overload responses changed from HTTP 503 to HTTP 429 with Retry-After.

BluePeak owns the client-side changes: bounded retries with exponential backoff and jitter, honouring Retry-After, and a client-side circuit breaker. Those are scheduled in the seller's October release train.

12. Validation

Check	Method	Result
Pool exhaustion under retry storm	Synthetic storm at 4x observed July peak, 24 Aug	Load shed at 429; no pool exhaustion
Recovery time after storm	Same test	Baseline within 40 seconds of storm ending

Check	Method	Result
Circuit breaker behaviour	Inventory lookup artificially slowed to 3 seconds	Breaker opened in 11 seconds; placements failed fast
Acquisition timeout	Pool saturated deliberately	Requests failed at 750 ms rather than queueing
No regression under normal load	72-hour production observation	p99 at 310 ms, unchanged

13. Preventive Actions

ID	Action	Owner	Target	Status
PA-1	Circuit breaker on inventory lookup	Vikram Singh	2026-08-26	Complete
PA-2	Pool resize and acquisition timeout	Vikram Singh	2026-08-26	Complete
PA-3	429 with Retry-After for overload	Vikram Singh	2026-08-26	Complete
PA-4	Requirement NFR-5: shed load gracefully	Engineering	2026-07-28	Complete
PA-5	10-working-day capacity notice, contractual	Neha Gupta	2026-07-28	Complete
PA-6	Capacity model for declared sale profiles	Neha Gupta	2026-10-09	In progress
PA-7	Client retry hardening at BluePeak	BluePeak	2026-10-23	Not started
PA-8	Dependency-degradation load test in the release pipeline	Engineering	2026-11-20	In progress

14. Follow-Up

The commercial follow-up was as significant as the technical one. The 10-working-day capacity notice is now a contractual expectation on this account rather than a courtesy, giving the capacity model a defined input.

Post-incident review held 18 July 2026 with Engineering, Support, Operations and the seller. The seller's initial position was that the platform had failed to handle its traffic; Engineering's was that the traffic had been amplified by the client. Both were accepted as true and the review closed with a joint workstream rather than an attribution of fault.